

DRISHTI: EVALUATOR DEFENSE HANDBOOK

Problem Statement: Early-Warning Fusion with Uncertainty-Aware Action Guidance

Live System: drishti-kohl.vercel.app

Prepared For: Jury Viva & Cross-Questions

 **Golden Rule for Tomorrow:** Evaluators love simple, confident answers with real-world examples. Speak like a problem solver, not a textbook. Keep answers to 2-3 clear sentences!

SECTION 1: "Why This Website?" vs Other Existing Platforms

Q1 "Why did you build this website? Platforms like SACHET (NDMA), 112 India, and Google Crisis Response already exist!"

 **How to Answer:**

"Sir, existing platforms only broadcast **one-way warning alerts** or show simple red dots on a map. They don't tell the user if an alert is 20% likely or 90% likely, nor do they give step-by-step guidance. **DRISHTI bridges this gap in 3 ways:** 1. **Uncertainty Quantification:** It separates high-certainty dangers (Immediate Evacuation) from low-certainty rumours (Dispatch Recon). 2. **Two-Way Field Verification:** It doesn't just push alerts; it lets citizens upload evidence that is cross-checked with real-time sensors. 3. **Dynamic Evacuation Navigation:** It automatically routes ambulances and victims away from flooded/blocked roads to open shelters."

Q2 "Why a Web Application (PWA) instead of an Android / iOS mobile app from Play Store?"

 **How to Answer:**

"Sir, during a sudden cyclone or flood, a panicking victim with 10% battery and weak 2G network **cannot wait to download a 60MB app from the Play Store** and go through OTP signup. Our web platform loads in **less than 1 second from a simple link or QR code**, works instantly without mandatory signup, and functions 100% offline as a Progressive Web App (PWA) using cached IndexedDB storage."

Platform	What They Do	What They Lack	How DRISHTI Wins
SACHET (NDMA)	One-way SMS/CAP broadcast	No citizen feedback, no dynamic routing	Two-way fusion with crowd consensus & bypass routing
112 India App	Manual police/SOS call dispatch	No sensor telemetry, no flood avoidance	Real-time USGS & weather correlation with auto-triage
Google Crisis Maps	Broad regional danger outlines	No uncertainty scoring, no responder triage	Bayesian 95% Credible Intervals & 3-tier action guidance
Twitter / Social Feeds	Rapid crowdsourced photos	High spam, fake news, panic rumours	4-tier anti-spam with Shannon Entropy image forensics

Q3 "Why would state disaster agencies like OSDMA or NDMA adopt your solution?"

How to Answer:

*"Sir, because DRISHTI complies with the official **ITU / NDMA Common Alerting Protocol (CAP v1.2)**. It does not replace government infrastructure; it acts as a lightweight, intelligent edge layer that filters out 90% of fake spam calls before they overwhelm emergency control room dispatchers."*

SECTION 2: Core Problem Statement (Uncertainty & Bayesian Fusion)

Q4 "What does 'Uncertainty-Aware Action Guidance' mean in simple terms?"

How to Answer:

"Sir, in simple words: **Do not treat a single tweet the same as verified sensor data**. • If only 1 person reports a flood with no rain sensors matching (High Uncertainty) → The system guides volunteers to **'Hold & Verify with Drone/Recon'**. • If multiple citizens report it and USGS/rain sensors spike (Low Uncertainty) → The system triggers **'Immediate Mass Evacuation & Red Alert'**. Action is strictly matched to confidence."

Q5 "Why did you choose Bayesian Beta-Binomial Updating instead of Deep Learning (PyTorch/YOLO)?"

How to Answer:

"Sir, deep neural networks are 'black boxes' that often make confident false predictions on new disaster scenarios where training data is scarce. Bayesian math works like a human doctor: it starts with a known baseline (prior sensors) and updates belief as new evidence arrives. It gives us a mathematically provable **95% Credible Interval** that emergency commanders can legally trust."

Q6 "How do you separate Epistemic Uncertainty from Aleatoric Noise?"

How to Answer:

"Sir, **Epistemic Uncertainty** is 'lack of information' (e.g. only 1 report from an isolated village)—we can reduce it by sending a volunteer or gathering more reports. **Aleatoric Noise** is natural physical fluctuation (e.g. GPS satellite drift or wind gusts)—it cannot be eliminated, so we build a safety buffer into our evacuation radius."

SECTION 3: Live Data, Anti-Spam & Map Routing

Q7 "Where is your live data coming from? Is it actually working right now?"

How to Answer:

"Yes sir, 100% live: 1. **Seismic Telemetry**: Fetched live from the **USGS Global Earthquake API** every 60 seconds. 2. **Weather & Rain**: Fetched live from **Open-Meteo**. 3. **Hospitals & Shelters**: Queried directly from **OpenStreetMap (Overpass API)** around the user's GPS."

Q8 "How do you detect fake flood reports or prank submissions?"

How to Answer:

"Sir, we use 4 automatic defense layers: 1. **Keyword Filter**: Flags pranks ('test', 'haha', 'fake'). 2. **Spatial Corroboration**: Requires cluster agreement from multiple phones within 2.5 km. 3. **Physical Sensor Cross-Check**: Checks if rain gauges in the area actually show precipitation. 4. **Image Forensics**: Calculates Shannon Byte Entropy and edge variance to reject blacked-out or stock images."

Q9 "What happens if a road is flooded on the map?"

How to Answer:

"Sir, verified hazard locations are marked as blocked obstacles. Our OSRM routing engine automatically steers ambulances and citizens around the obstacle to the safest nearby shelter."

SECTION 4: Offline Resilience, Blackouts & Scalability

Q10 "What happens during a cyclone when cellular towers collapse and there is NO internet?"

 **How to Answer:**

"Sir, DRISHTI is an **Offline-First PWA**. The map tiles, hospital directory, and emergency guides are stored locally in the browser's **IndexedDB**. Users can still create incident reports offline (**PendingSync** status). These are shared over local browser mesh (**BroadcastChannel**) and auto-sync upstream the instant any network connection returns."

Q11 "How will your server handle 10 Lakh (1 Million) concurrent visitors during a disaster?"

 **How to Answer:**

"Sir, DRISHTI uses an **Edge-Computing Architecture**. The AI Bayesian checks and route calculations run directly on the user's phone, not our central server. Static assets are cached across global Edge CDNs, keeping server load near zero."

Q12 "How do illiterate or elderly citizens use this platform in high panic?"

 **How to Answer:**

"Sir, our Quick Emergency Help screen has **3 massive, high-contrast action buttons** (**Red Police 112**, **Green Ambulance 108**, **Orange Fire 101**) with 1-tap direct calling and simple visual icons for trapped/medical help, requiring zero text typing."

SECTION 5: Future Roadmap & Limitations

Q13 "What are the current limitations of your prototype, and what is your future roadmap?"

 **How to Answer:**

"Sir, currently we rely on public sensor APIs (USGS/Open-Meteo) and browser storage. **Our Next Phase includes:** 1. Direct integration with India's C-DOT SACHET alerting gateway. 2. Bluetooth Low Energy (BLE) peer-to-peer mobile phone relay for 100% off-grid mesh rescue. 3. Drone (UAV) orthomosaic video stream overlays directly on the tactical map."

Q14 "Give me your 15-second elevator pitch."

 **How to Answer:**

"Respected Evaluators, in disasters like the Wayanad landslides, the problem is not a lack of data—it is data conflict and panic. **DRISHTI** fuses live sensors with citizen reports, mathematically calculates the uncertainty, and guides responders and citizens with clear, life-saving actions in real time."